

MULTI-VLAN ENTERPRISE NETWORK

Project Documentation

A Packet Tracer network implementing VLAN segmentation, OSPF dynamic routing, centralized DHCP and DNS services, DHCP relay, and secure remote management using SSH.

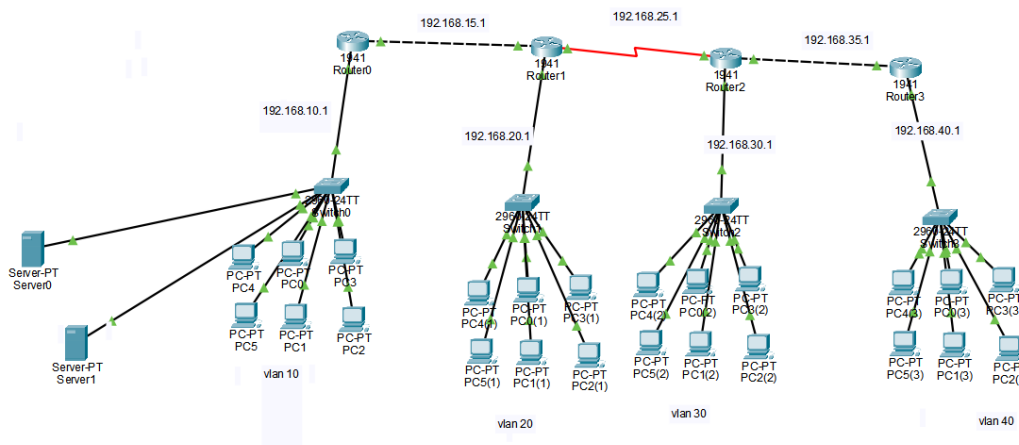


Figure 1. Implemented Packet Tracer topology.

Network Features

VLAN 10 • VLAN 20 • VLAN 30 • VLAN 40

OSPF • Centralized DHCP • DNS • DHCP Relay • SSH Version 2

1. Project Overview

This project implements a multi-segment enterprise network in Cisco Packet Tracer. The network is divided into four logical LANs using separate IP networks referred to in the project as VLAN 10, VLAN 20, VLAN 30, and VLAN 40. Each LAN has its own router interface acting as the default gateway. Four routers are interconnected and use OSPF to exchange routing information. DHCP and DNS are centralized on servers located in the VLAN 10 server segment.

2. Project Objectives

Objective	Implementation
Network segmentation	Four isolated LAN/VLAN segments using separate IPv4 /24 networks.
Dynamic routing	OSPF area 0 exchanges routes between the four routers.
Automatic addressing	One centralized DHCP server provides addresses to all four VLANs.
Cross-subnet DHCP	Routers use DHCP relay (ip helper-address) for remote VLANs.
Name resolution	A centralized DNS server is deployed in VLAN 10.
Secure management	SSH Version 2 with local authentication is enabled on routers.

3. Network Topology

The topology is built around four routers connected in sequence. Each router provides connectivity for one user LAN. Router0 hosts the server-side LAN containing the centralized DHCP and DNS servers. Router-to-router transit networks provide end-to-end connectivity, while OSPF distributes reachability information automatically.

Logical Design

DHCP Server (192.168.10.10) and DNS Server (192.168.10.11) → VLAN 10 → Router0 → Router1 → Router2 → Router3 → VLAN 40, with VLAN 20 and VLAN 30 connected to Router1 and Router2 respectively.

4. IP Addressing Plan

Segment	Network	Default Gateway / Router IP	Notes
VLAN 10	192.168.10.0/24	192.168.10.1	Server segment; DHCP and DNS located here
VLAN 20	192.168.20.0/24	192.168.20.1	Client LAN
VLAN 30	192.168.30.0/24	192.168.30.1	Client LAN
VLAN 40	192.168.40.0/24	192.168.40.1	Client LAN
R0–R1 Transit	192.168.15.0/24	Router interfaces	Inter-router link
R1–R2 Transit	192.168.25.0/24	Router interfaces	Inter-router link
R2–R3 Transit	192.168.35.0/24	Router interfaces	Inter-router link

5. Network Segmentation

The four user/server segments use different IPv4 networks. This separation reduces broadcast scope and provides a structured foundation for future access-control policies. Devices in each segment use the corresponding .1 address as their default gateway.

6. OSPF Dynamic Routing

OSPF process 1 is configured in area 0. Each router advertises its directly connected LAN and transit networks. This allows a device in one VLAN to reach devices and services located behind the other routers without manually configuring static routes for every destination.

Example OSPF configuration pattern

```
router ospf 1
network <connected-network> 0.0.0.255 area 0
network <transit-network> 0.0.0.255 area 0
```

Example observed on the VLAN 30 router: the networks 192.168.30.0/24, 192.168.25.0/24, and 192.168.35.0/24 are advertised into OSPF area 0.

7. Centralized DHCP Service

A single DHCP server is deployed in VLAN 10 and provides dynamic IPv4 addressing for all four networks. Centralization simplifies administration because address pools are managed from one location instead of configuring a separate DHCP service on every router.

DHCP Server Setting	Value
IPv4 Address	192.168.10.10
Subnet Mask	255.255.255.0
Default Gateway	192.168.10.1
DNS Server	192.168.10.11

DHCP Pools

Pool	Network	Gateway	DNS	Starting Address
VLAN10_Pool	192.168.10.0/24	192.168.10.1	192.168.10.11	Configured pool start
VLAN20_Pool	192.168.20.0/24	192.168.20.1	192.168.10.11	Configured pool start
VLAN30_Pool	192.168.30.0/24	192.168.30.1	192.168.10.11	Configured pool start
VLAN40_Pool	192.168.40.0/24	192.168.40.1	192.168.10.11	Configured pool start

8. DHCP Relay (ip helper-address)

DHCP Discover messages are broadcasts and normally do not cross routers. Therefore, clients in VLAN 20, VLAN 30, and VLAN 40 cannot directly broadcast to the DHCP server in VLAN 10. The router interface facing each remote client network is configured with an IP helper address that forwards DHCP requests to the centralized DHCP server at 192.168.10.10.

```
interface <LAN-interface-facing-the-client-VLAN>
ip helper-address 192.168.10.10
```

DHCP operation: Client → DHCP Discover → Local Router → ip helper-address relay → 192.168.10.10 DHCP Server → DHCP Offer/Acknowledgment → Client.

9. DNS Service

A dedicated DNS server is placed in VLAN 10 with address 192.168.10.11 and default gateway 192.168.10.1. DHCP clients receive this DNS address automatically, allowing centralized name resolution across all network segments.

10. Secure Remote Management Using SSH

SSH Version 2 is enabled on all four routers. Unlike Telnet, SSH encrypts the management session. Local authentication is used and the VTY lines are restricted to SSH input, preventing Telnet access.

Security-related configuration pattern

```
ip domain-name network.local
username <admin-user> privilege 15 secret <password>
crypto key generate rsa
ip ssh version 2

line vty 0 4
login local
transport input ssh
```

Verification command

```
show ip ssh
```

The routers were verified to display: “SSH Enabled - version 2.0”. The VTY configuration was also verified with “login local” and “transport input ssh”, confirming that remote VTY access is restricted to SSH.

11. Testing and Verification

Test	Expected Result
DHCP acquisition	Clients receive an IPv4 address, /24 mask, correct default gateway, and DNS server.
Inter-VLAN ping	A client can ping reachable hosts in another VLAN through OSPF routing.
Server reachability	Clients can reach the centralized DHCP/DNS server network.
Packet simulation	ICMP packets may be inspected in Simulation Mode to observe forwarding behavior.
SSH verification	show ip ssh reports SSH Enabled - version 2.0.
VTY verification	show running-config section line vty shows login local and transport input ssh.

12. Security Considerations and Future Improvements

The current implementation already includes secure router administration through SSH Version 2. The following enhancements can be added in a future version after backing up the Packet Tracer file:

- Access Control Lists (ACLs) to restrict communication between selected VLANs.
- Port Security on access ports only, preferably using a non-disruptive violation mode such as restrict.
- BPDU Guard and PortFast on end-device access ports.
- Shutdown of unused switch ports after confirming which interfaces are not in use.
- Syslog server for centralized logging and monitoring.
- Password encryption and an unauthorized-access MOTD banner.

13. Conclusion

The project successfully demonstrates an integrated enterprise-style network using four segmented LANs, OSPF dynamic routing, centralized DHCP and DNS services, DHCP relay for cross-router address assignment, and SSH Version 2 for secure router administration. The design provides a clear separation of network functions while maintaining end-to-end connectivity. Centralizing DHCP and DNS reduces administrative overhead, and OSPF provides automatic route distribution as the topology expands.

Final Implemented Feature Summary

Feature	Status
Four network segments: VLAN 10/20/30/40	Implemented
OSPF dynamic routing	Implemented
Central DHCP server at 192.168.10.10	Implemented
DHCP pools for four VLANs	Implemented
DHCP relay using ip helper-address	Implemented
DNS server at 192.168.10.11	Implemented
SSH Version 2 on routers	Implemented and verified
Telnet blocked on VTY lines	Implemented via transport input ssh